

Coding exercise 1: pricing a chocolate brand with linear regression

Individual · 5 % of the final grade · International Marketing Analytics · WU Vienna · WT 2026/27

! In brief

Work	individual
Weight	5 % of the final grade (20 points)
Submit	<code>exercise1_lastname.qmd</code> and the rendered <code>.html</code> on Canvas
Deadline	see Canvas
Data	<code>data/legacy/chocolate_dataset.csv</code> in the course repository: 68 weeks of real scanner data, unit sales of brand 1 and prices, features and displays of four chocolate brands
Tools	polars, plotnine, great_tables and statsmodels

The code from Session 1 (`sessions/01-foundations/regression_chocolate.qmd`) is your starting point.

Packages

Install once (skip if you did it in class): pick **Python 3.14 (.venv)** at the top right of Positron and run this line in a cell or the Console. In your submitted file, keep it in a cell with `#| eval: false`, so that Quarto does not run it when rendering.

```
%pip install polars plotnine great-tables statsmodels pyarrow jupyter
```

Load at the top of your file, in the first code cell:

```
import numpy as np                # np.log in model formulas
import polars as pl               # data
import statsmodels.formula.api as smf  # regression
from great_tables import GT       # tables
from plotnine import (aes, geom_histogram, geom_hline,    # charts
                      geom_line, geom_point, ggplot, labs)
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.stats.stattools import durbin_watson, jarque_bera
```

pyarrow and jupyter are never imported, but must be installed: pyarrow passes polars data to plotnine and statsmodels, jupyter lets Quarto run Python. Install `great-tables`, load `great_tables`. Error *No module named ...?* See Step 0 of `regression_chocolate.qmd`. This exercise uses a different model and a different cost, so the answers differ from the in-class results.

Tasks

- 1. **Explore.** A `great_tables` table with the mean, minimum and maximum of `sales`, `price1` to `price4`, `feature1` and `display1`. Two plotnine charts: sales over the 68 weeks, and sales against `price1`. Describe in two sentences what you see.
- 2. **Simple model.** Estimate `sales ~ price1`. Interpret the slope for a price change of 10 cents and report R^2 .
- 3. **Log-log model.** Estimate `np.log(sales) ~ np.log(price1) + np.log(price2) + np.log(price3) + np.log(price4) + feature1 + display1 + temp + december`. Interpret the own-price elasticity, one competitor price and `december` (use $\exp(b) - 1$ for the dummy). Which coefficients are significant at 5 %?
- 4. **Assumptions.** A plot of residuals against fitted values, a histogram of the residuals, the Durbin-Watson statistic, the Breusch-Pagan and Jarque-Bera p-values and the VIFs, in one table. One sentence per check: is the assumption met, and what does it mean for your results? Comment on the VIF of `display1`.
- 5. **Robust errors.** Refit the model with Newey-West standard errors (`model.model.fit(cov_type="HAC", cov_kwds={"maxlags": 1})`). What changes, and what does not?
- 6. **Recommendation.** With a marginal cost of 0.90, compute the price that the own-price elasticity suggests ($p^* = \text{cost} \times \epsilon / (1 + \epsilon)$). In at most 100 words, recommend a regular price to the brand manager and give two caveats.

Every number in your text must come from code (inline code such as ``{python} f" {eps:.2f}"``). End the file with a **prompt log**: the prompts you used with an AI assistant, and what you kept or changed.

Grading (20 points)

Criterion	Points
Code runs from top to bottom; charts and tables clean and labelled	5
Interpretation of the coefficients (units, per cent, significance)	6

Criterion	Points
Assumption checks and a sound conclusion	5
Recommendation with the computed price and two caveats	2
Prompt log	2
Total	20

Individual work: discuss ideas, but write and check your own code. AI assistants are allowed; you own every number. Questions go to the Q&A forum on Canvas.